

NATIONAL JUNIOR COLLEGE, SINGAPORE
Senior High 2
Preliminary Examination
Higher 2

CANDIDATE
NAME

BIOLOGY
CLASS

2bi2____ / 2IPbi2__

REGISTRATION NUMBER

BIOLOGY

Paper 2

9648/02

26 August 2016

2 hours

Additional Materials: Answer Paper

READ THESE INSTRUCTIONS FIRST

Write your name and Biology class on all the work you hand in.
Write in dark blue or black pen.
You may use an HB pencil for any diagrams or graphs.
Do not use staples, paper clips, highlighters, glue or correction fluid.

Sections A - D

Answer **all** questions in the spaces provided on the question paper.

Section E

Answer any **one** question on the answer paper provided.

The use of an approved scientific calculator is expected, where appropriate.

You may lose marks if you do not show your working or if you do not use appropriate units.

At the end of the examination, fasten all your work securely together.

The number of marks is given in the brackets [] at the end of each question or part question.

For Examiner's Use	
Section A	(Total: 23)
1	/ 12
2	/ 11
Section B	(Total: 19)
3	/ 7
4	/ 12
Section C	(Total: 19)
5	/ 12
6	/ 7
Section D	(Total: 19)
7	/ 7
8	/ 12
Section E	(Total: 20)
9 or 10	/ 20
Total	/ 100

This document consists of **24** printed pages.

Section A

Answer **all** the questions in this section.

- 1** Many bacteria can digest cellulose using a group of enzymes called cellulases. Cellulases **A** and **T** were extracted from two different bacteria, *Agrobacterium tumefaciens* and *Thermotoga maritima*, respectively.

Fig. 1.1 shows the results of an investigation into the effect of temperature on the activity of each enzyme.

L represents the lowest temperature at which activity of each enzyme was detected.
H represents the highest temperature at which activity of each enzyme was detected.

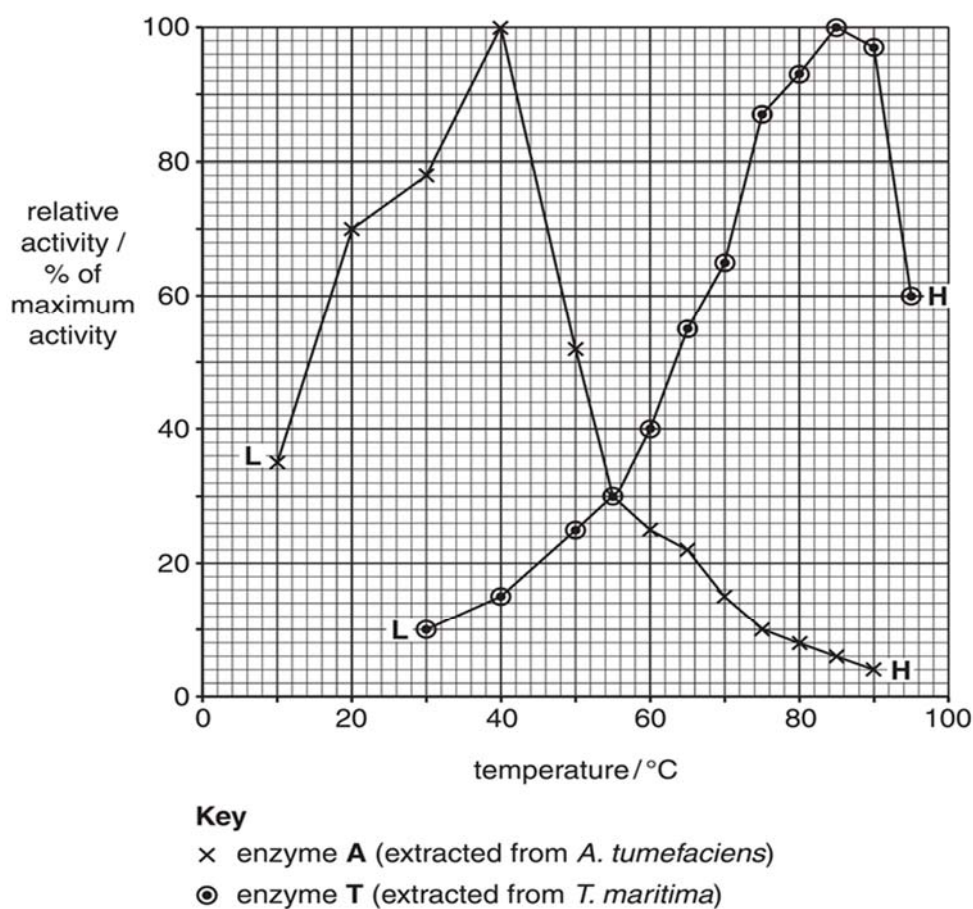


Fig.1.1

- (a) With reference to Fig. 1.1, describe two differences in the results for the two enzymes, A and T. [2]

[2]

- (b)** With reference to Fig. 1.1, explain the effect of increasing temperature on the relative activity of enzyme **T**.

----- [3]

- (c)** Suggest a structural feature of enzyme **T**, which helps to explain the results obtained in the investigation.

----- [2]

Fig. 1.2 shows the structure of small sections of DNA and messenger RNA (mRNA) in the nucleus of *Agrobacterium tumefaciens* during transcription of the gene coding for cellulase.

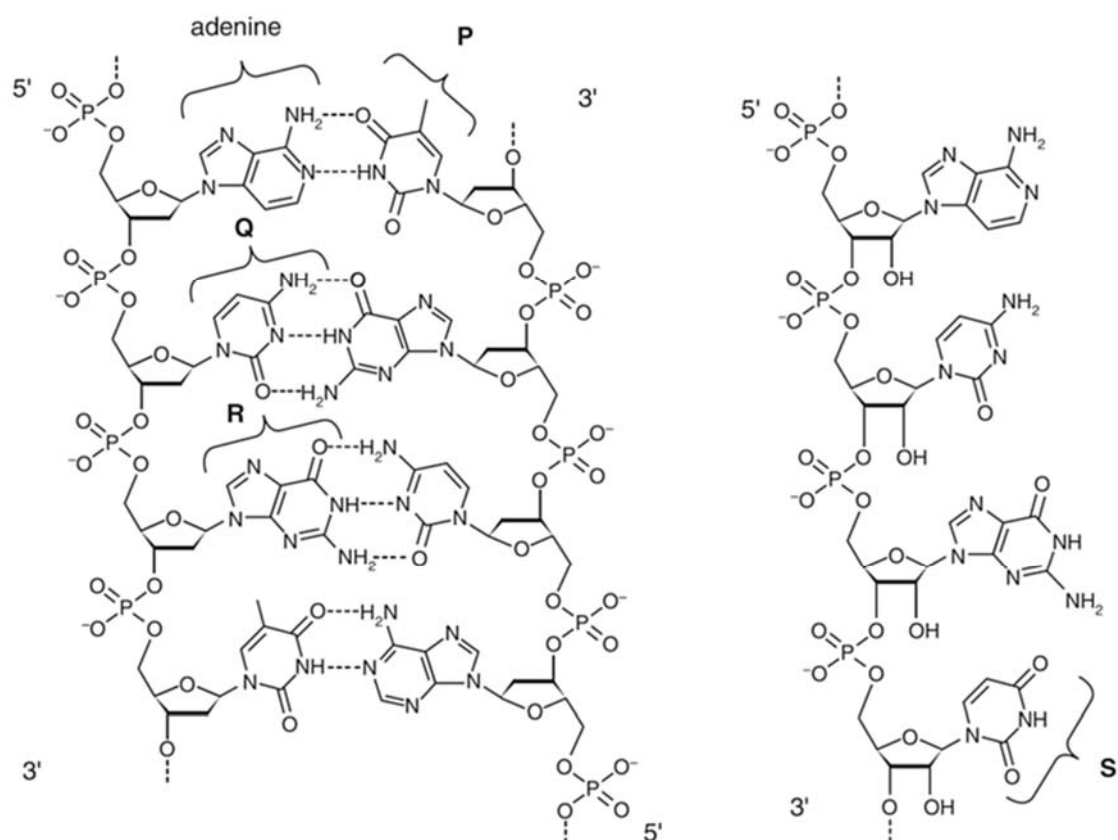


Fig. 1.2

(d) Name the bases P to S.

P:

Q:

R:

S:

[2]

- (e) Describe how messenger RNA coding for cellulase is synthesised in *Agrobacterium tumefaciens*.

----- [3]

[Total: 12]

2

Fig. 2.1 shows the main steps involved in the synthesis of preproinsulin to insulin in the pancreatic β -cell. The preproinsulin is synthesised into the lumen of organelle **A** as proinsulin.

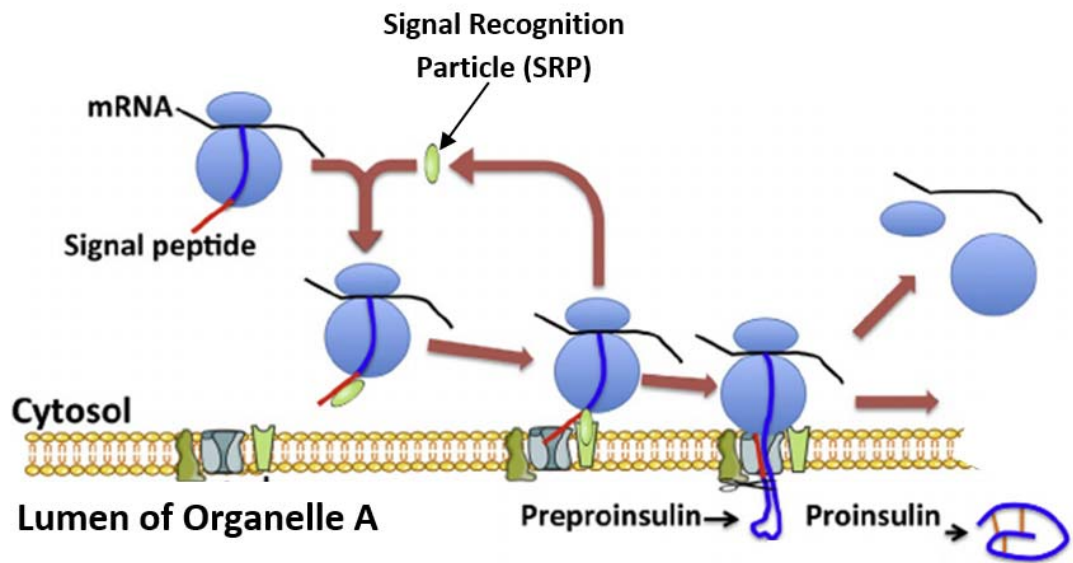


Fig. 2.1

The proinsulin is then be transported to organelle **B** where it is further processed to form insulin.

Fig. 2.2 shows the conversion of proinsulin to insulin in organelle **B**.

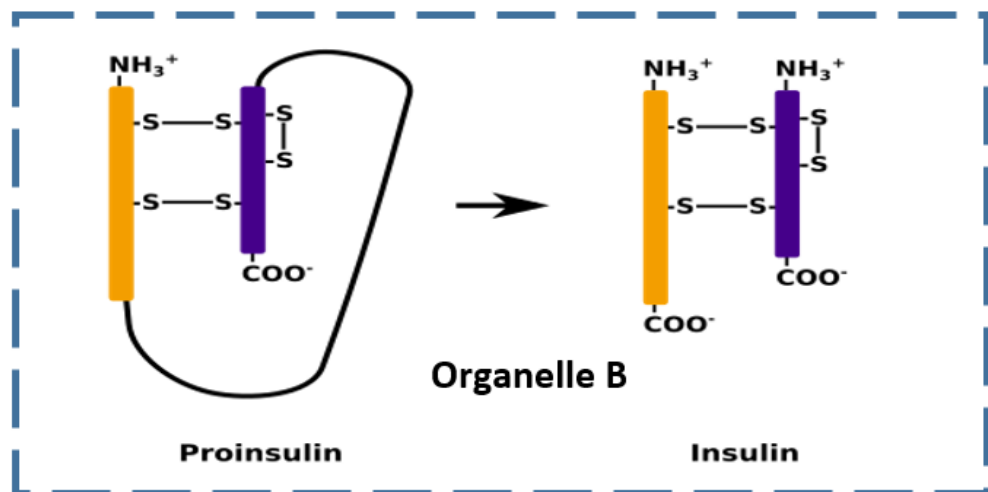


Fig. 2.2

(a) Name the organelles labelled **A** and **B**.

organelle **A**:

organelle **B**:

[1]

(b) State the role of rRNA in insulin synthesis.

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[2]

(c) Insulin is released by pancreatic β -cell. Outline the route taken by proinsulin.

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[2]

The synthesis of insulin is regulated at the transcriptional levels in β -cell.

Fig. 2.3 shows the binding of three β -cell-specific transcriptional factors, Pdx-1, MafA and NeuroD1 E47 in response to high glucose levels in β -cell.

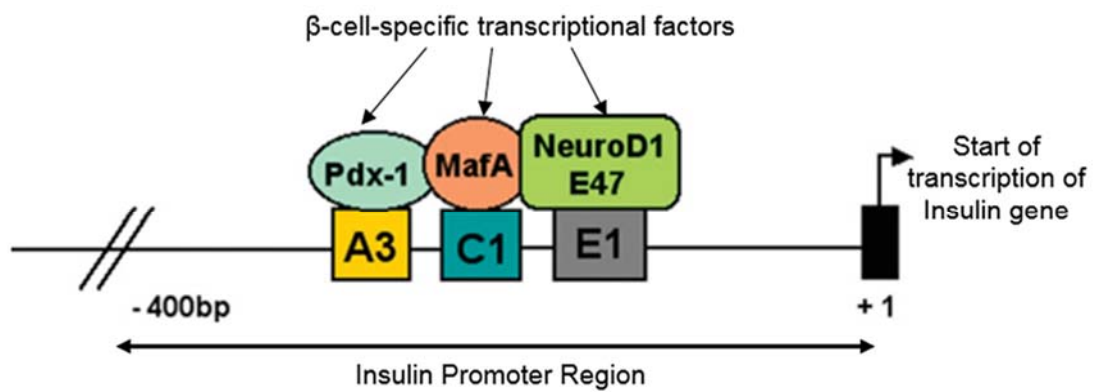


Fig. 2.3

- (d) Explain how β -cell-specific transcriptional factors will lead to high level of insulin gene expression.

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..... [2]

- (e) It has been found that insulin promoters differ widely in efficiency among individuals. Strong promoters cause frequent initiations of transcription while weak promoters have low frequency of initiations of transcription. Explain what may have caused the difference in efficiency of the insulin promoters.

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..... [2]

- (f) Explain why gene mutations do not always produce mutated insulin protein whereas mutations of the splicing sites involved in RNA splicing will produce mutated insulin.

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..... [2]

[Total: 11]

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Section B

Answer **all** the questions in this section.

- 3 Fig. 3.1 shows the life cycle of a water flea.

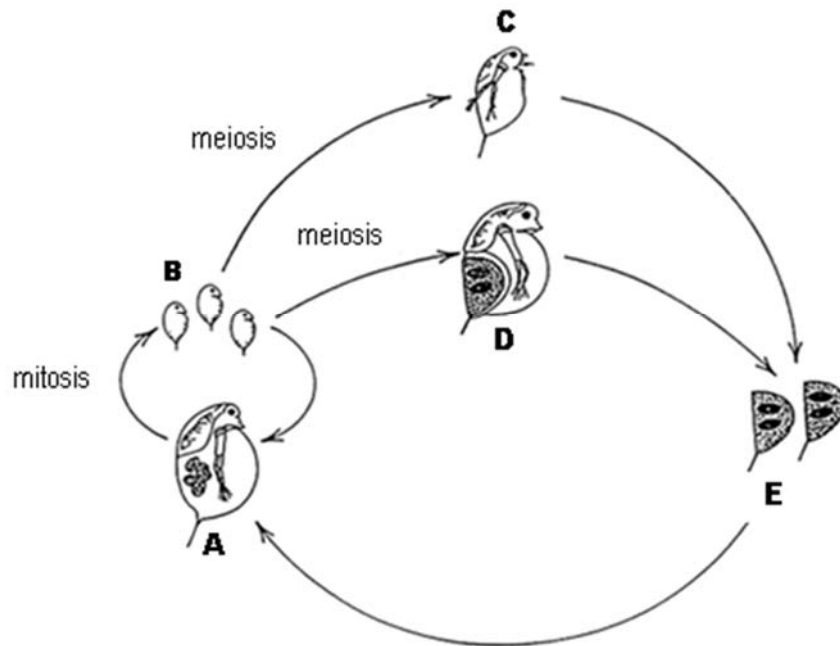


Fig. 3.1

In favourable conditions, all the animals in a population are females (**A**). These females produce eggs by mitosis, which develop into young females (**B**) without being fertilized. In unfavourable conditions, eggs produced by meiosis develop directly without fertilization into either males (**C**) or females (**D**). The eggs produced by the females (**D**) are fertilized by the sperms from the males (**C**), then released in protective egg cases (**E**) which enable them to survive unfavourable conditions. When favourable conditions return, these eggs develop back into females (**A**).

- (a) The females at stage **A** of the life cycle have 18 chromosomes.

Complete the table to show the number of chromosomes at the other stages of the life cycle.

stage of life cycle	chromosome number
A	18
B	
C	
D	
E	

[1]

- (b) Explain why the eggs from **D** and the sperms from **C** must be produced by mitosis.

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..... [2]

- (c) Explain why females **A**, developed from fertilized eggs **E**, are genetically different from each other.

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..... [3]

- (d) Give an example of a favourable condition in which females will develop from eggs formed via mitosis.

..... [1]

[Total: 7]

- 4 Cattle were found to have three different coat colours: brown, white and roan. Roan cattle have both brown and white patches. When two roan cattle were crossed, a ratio of 1:2:1 was obtained for brown, roan and white coat colour respectively. A separate gene on X chromosome in cattle code for a disease called Agnathia. The absence of a normal allele causes a deformity in the lower jaw.

- (a) Construct a genetic diagram to show the expected ratio when brown carrier cows are crossed with normal jawed white bulls.

[5]

- (b) State the mode of inheritance of the two traits. Explain how you arrive at your answer.

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[2]

- (c) When a geneticist carried out a cross between brown carrier cows and normal jawed white bulls, he obtained the following phenotypes.

normal jawed, roan cows	55
normal jawed, roan bulls	37
deformed jawed, roan bulls	28

Fig. 4.1 shows the table of probabilities.

df	probability				
	0.10	0.05	0.02	0.01	0.001
1	2.71	3.84	5.41	6.64	10.83
2	4.61	5.99	7.82	9.21	13.82
3	6.25	7.82	9.84	11.35	16.27
4	7.78	9.49	11.69	13.28	18.47
5	9.24	11.07	13.39	15.09	20.52

Fig. 4.1

The formula for the chi-square statistic used in the chi square test is as follows:

$$\chi^2 = \sum \frac{(O - E)^2}{E} \quad \text{where } \begin{array}{l} O = \text{observed value;} \\ E = \text{expected value.} \end{array}$$

With reference to Fig. 4.1, carry out a chi-square test to support your explanation in (b). Show your workings clearly in the space below.

[3]

(d) In some cases, the environment may affect the phenotype.

Give one named example to illustrate such an environmental effect.

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[2]

[Total: 12]

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Section C

Answer **all** the questions in this section.

- 5 Hepatitis C virus (HCV) is an enveloped, positive-strand RNA virus within the family Flaviviridae. HCV undergoes reproductive cycles similar to other enveloped viruses, such as influenza virus and dengue virus.

Fig. 5.1 shows a model of a HCV particle.

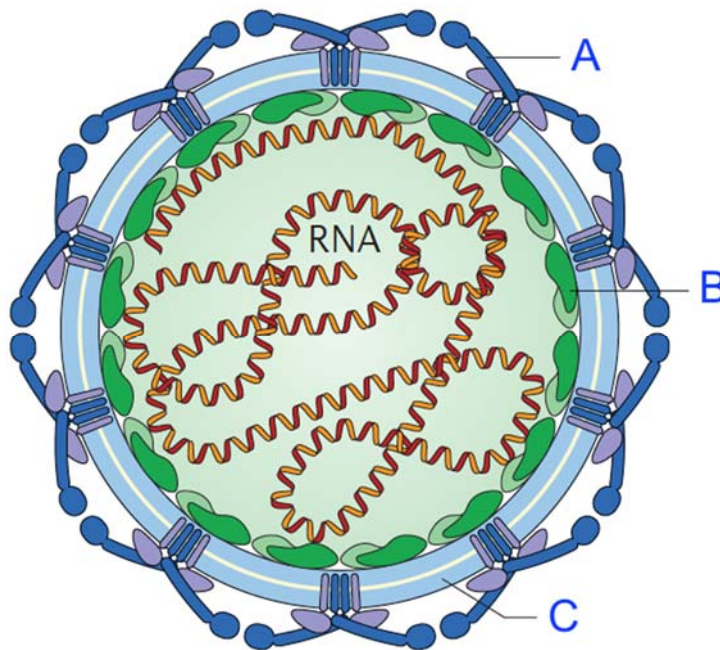


Fig. 5.1

- (a) Identify the labelled structures A, B and C.

A:

B:

C:

[2]

- (b) Explain why HCV is considered an obligate parasite.

[2]

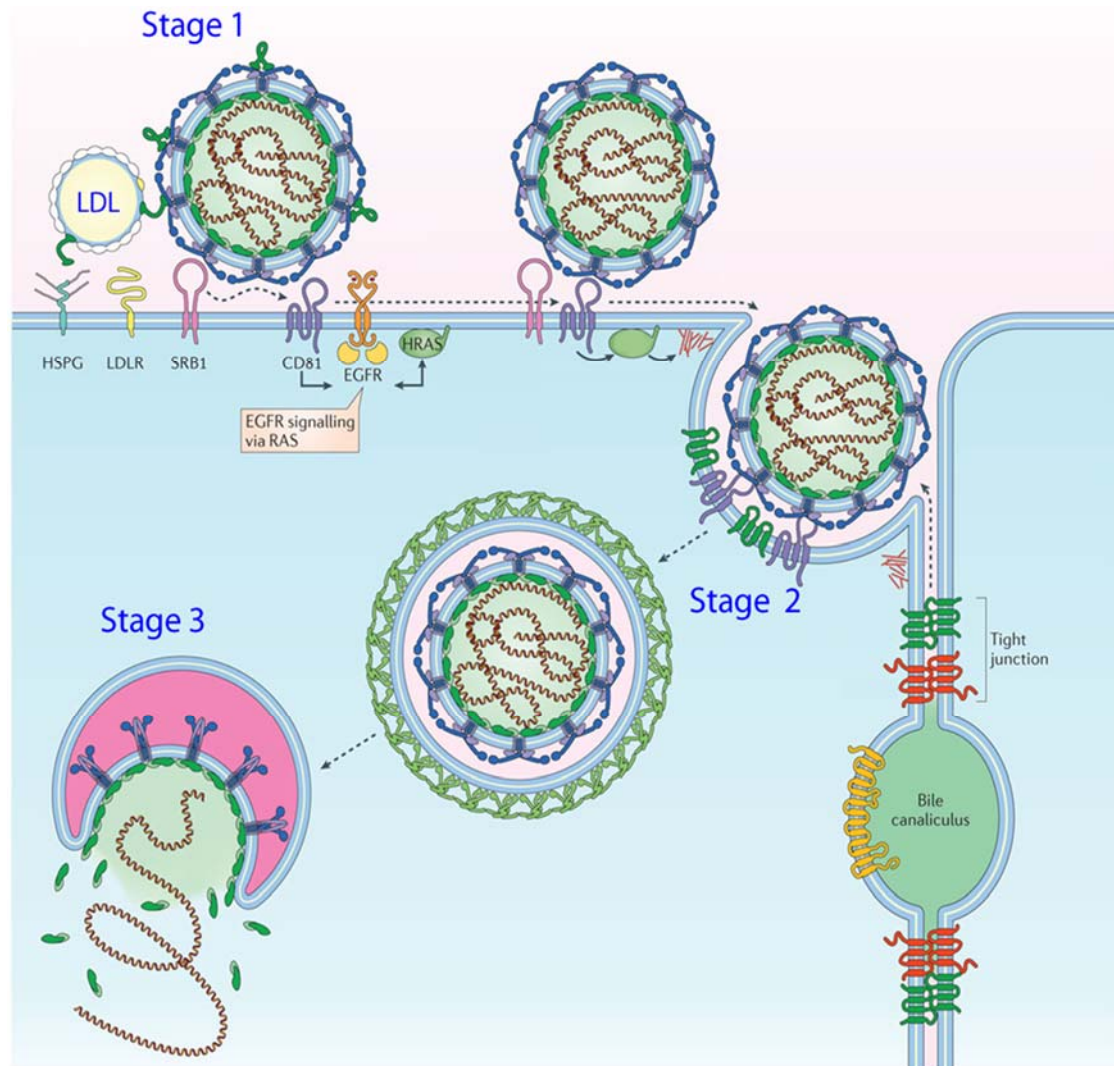


Fig. 5.2

- (c) With reference to all labelled stages in Fig. 5.2, describe how HCV gains entry into its host cell.

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[3]

Fig. 5.3 shows the positive-strand RNA genome of HCV genome.

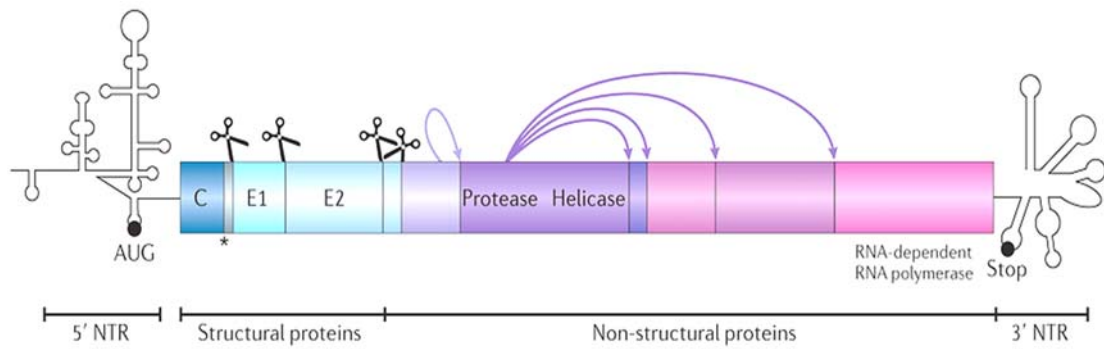


Fig. 5.3

(d) With reference to Fig. 5.3, describe the role of HCV genome in the infection process.

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[2]

Hepatitis B virus (HBV) and HCV are known to be two major causative agents of hepatocellular carcinoma. Studies have shown that HBV DNA can be integrated during the early stages of infection. The integration of viral DNA is associated with deletions in portions of the host chromosomes. Many of these chromosomal segments contain known genes such as p53.

(e) Explain how HBV infection may lead to cancer.

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[3]

[Total: 12]

- 6 (a) Human newborns and hibernating mammals contain large amounts of brown adipose tissue ('body fat').

Fig 6.1 shows the electron micrograph of a brown adipocyte. Brown adipocytes are characterised by the presence of numerous vacuoles and organelle X throughout the cell.

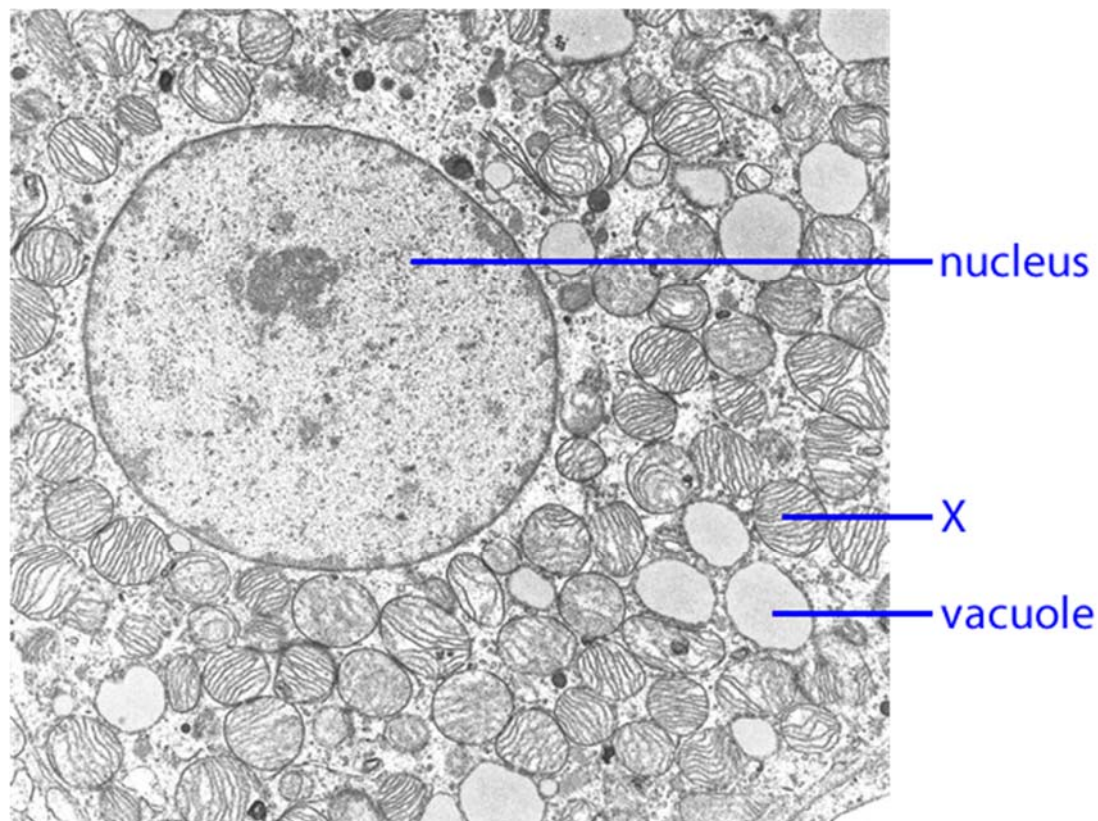


Fig. 6.1

- (i) Identify organelle X.

..... [1]

- (ii) Suggest the role of the numerous vacuoles found in brown adipocytes.

..... [1]

- (b) Fig. 6.2 shows the schematic representation of a series of protein complexes found on the inner membrane of organelle X.

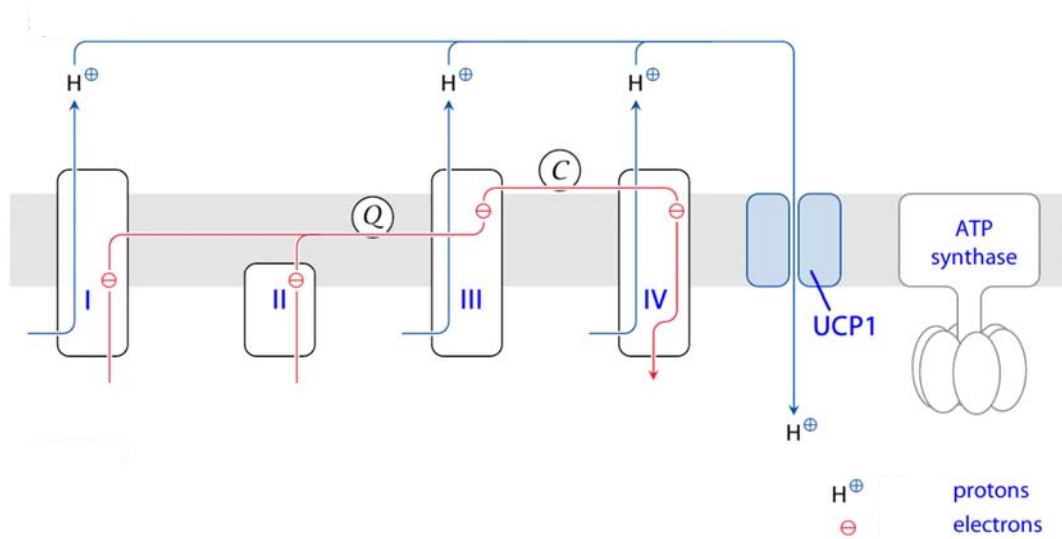


Fig. 6.2

- (i) Oxygen is required to sustain the process illustrated in Fig. 6.2. With reference to Fig. 6.2, describe the role played by oxygen.

..... [1]

- (ii) Brown adipocytes contain a unique protein, UCP1, which is not found in organelle X in any other cell type.

Evaluate the impact of UCP1 on the normal functioning of the process illustrated in Fig. 6.2 and suggest the physiological significance of brown adipose tissue.

..... [2]

- (c) In other cell types, NADH and FADH₂ are used to drive ATP synthesis by ATP synthase. Using relevant information from Fig. 6.2, suggest and explain why more ATP is produced from NADH.

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..... [2]

[Total: 7]

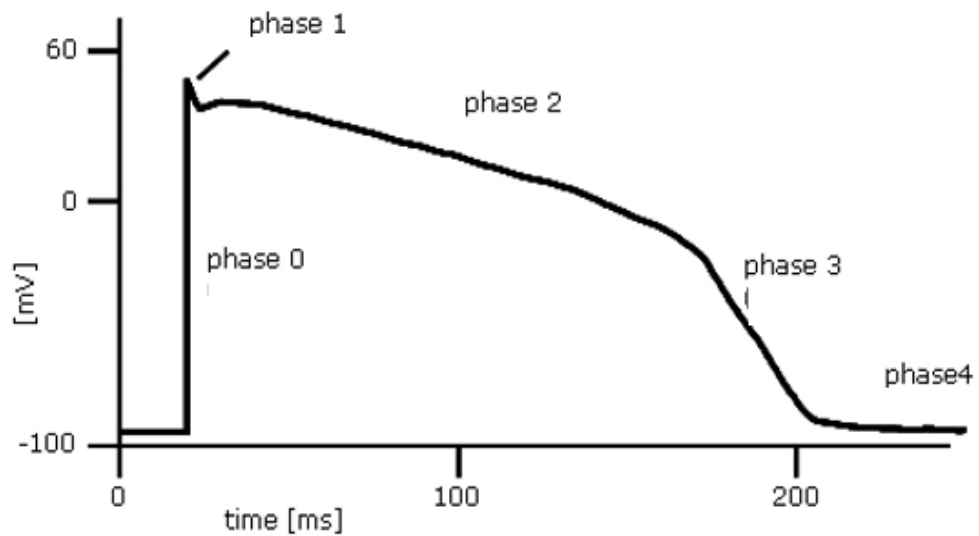
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Section DAnswer **all** the questions in this section.

- 7 The cardiac action potential is a specialised action potential with unique properties necessary for the electrical conduction system of the heart. The cardiac muscle cells share many things in common with nerve cells.

Fig. 7.1 shows the five phases of a cardiac action potential. Each of which is characterised with their respective changes in membrane potential.

**Fig. 7.1**

The table shows the relative concentrations of ions inside and outside of the cardiac muscle cells.

element	ion	extracellular	intracellular
sodium	Na ⁺	135 – 145	10
potassium	K ⁺	3.5 – 5.0	155
chloride	Cl ⁻	95 – 110	20 – 30
calcium	Ca ²⁺	2	10 ⁻⁴

- (a) Estimate the resting potential for the cardiac action potential shown in Fig. 7.1.

[1]

- (b) Using the information from the table, explain how the resting potential can be produced in the heart muscles using sodium and potassium ions.

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..... [3]

- (c) Compare phase 2 of the action potential of a cardiac muscle cell with that of a nerve cell.

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..... [1]

- (d) Cardiac arrhythmia refers to any abnormal electrical activity in the heart. As a result, the heart may beat too fast. Calcium channel blockers such as Verapamil are often used to treat this condition.

Suggest and explain the action of Verapamil in controlling this symptom of arrhythmia.

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..... [2]

[Total: 7]

- 8 The table shows the amino acid differences in the cytochrome b protein between various vertebrates.

	Human	Elephant	Platypus	Ostrich	Starling	Crocodile	Lungfish	Coelacanth	Goldfish	Shark
Human		26	40	43	41	47	83	70	68	71
Elephant			45	45	48	50	84	72	63	74
Platypus				54	52	51	89	74	70	76
Ostrich					26	36	91	75	68	73
Starling						47	91	77	67	70
Crocodile							85	78	70	77
Lungfish								90	94	86
Coelacanth									83	78
Goldfish										88
Shark										

Fig. 8.1 shows the phylogenetic tree based on differences between the cytochrome b proteins.

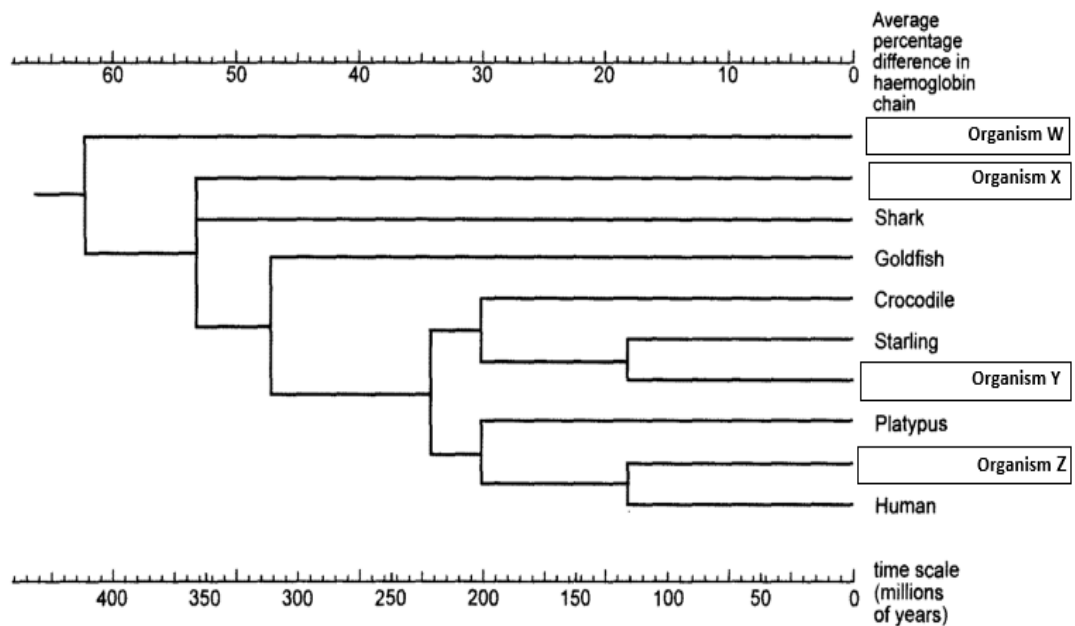


Fig. 8.1

- (a) Using information from the table and Fig. 8.1, identify organisms W to Z.

W:

X:

Y:

Z:

[2]

- (b) Explain how differences in amino acid sequences in the cytochrome b chain allow the establishment of the phylogenetic tree.

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..... [3]

- (c) Explain the difference between classification and phylogeny.

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- (d) Suggest why homology still features prominently in evolutionary studies despite the advantages that molecular evidence can confer.

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..... [2]

- (e) Explain the role of neutral mutations in evolutionary studies.

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..... [3]

[Total: 12]

Section E

Answer **one** question.

Write your answers on the separate answer paper provided.

Your answers should be illustrated by large, clearly labelled diagrams, where appropriate.

Your answers must be in continuous prose, where appropriate.

Your answers must be set out in sections **(a)**, **(b)** etc., as indicated in the question.

- 9 (a)** Compare the role of nervous system and endocrine system as communication systems within organisms. [6]
- (b)** Explain the meaning of the term homeostasis with specific reference to the control of raised blood glucose concentration in humans. [8]
- (c)** Describe the cell signalling pathway that glucagon initiates in order to regulate blood glucose concentration. [6]

[Total: 20]

- 10 (a)** Compare the structural and regulatory genes in prokaryotes. [6]
- (b)** Explain the roles of the operator and activator binding site in the *lac* operon. [8]
- (c)** Describe how the molecular structure of phospholipids is related to their function in the plasma membrane. [6]

[Total: 20]

- End of paper -